Yu Yuan

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Research Interests

Physics-Consistent and Controllable Video Generation, Physical World Models, Dynamics Modeling, 3D/4D Generation, Computational Photography, Image/Video Editing, Image/Video Enhancement, VLM.

Education

Purdue University West Lafayette, IN, US
PhD Student in Electrical and Computer Engineering Advisor: Stanley H. Chan *08/2023 – 02/2027 (expected)*

Shanghai Jiao Tong University Shanghai, China
MS in Aeronautical and Astronautical Science and Technology *09/2020 – 06/2023*

Shanghai Jiao Tong University Shanghai, China
BEng in Aeronautics and Astronautics Engineering *09/2016 – 06/2020*
Minor Degree in Administration Management *10/2018 – 06/2020*

Experience

Adobe Research San Jose, CA
Research Intern (Multimodal Foundation Models Team) *05/2026 – 08/2026*

- **Developed** a general-purpose framework for **causal video dynamics editing**, enabling counterfactual changes to actions, interactions, and motion while preserving visual context, identity, and scene appearance.
- **Introduced** a decoupled dynamics representation that compresses V-JEPA spatiotemporal features into editable dynamics tokens, together with a dynamics-conditioned Wan2.2 renderer and a latent-space editor for instruction-guided motion transformation.
- **Built** an end-to-end research pipeline spanning **10K+ paired counterfactual video edits**, large-scale data generation, distributed training, LoRA fine-tuning, benchmark construction, and systematic studies of dynamics controllability and editing.

Purdue University West Lafayette, IN
Research Assistant (Intelligent Imaging Lab) *08/2023 – Present*

- **Led** research on physics-aware generative AI, controllable video generation, 4D world modeling, and computational photography, incorporating physical dynamics, optimal control, and continuous spatiotemporal representations into generative models.
- **Authored 4 first-author papers** at top-tier venues including **CVPR 2025 Highlight, ICLR 2026, CVPR 2026, and NeurIPS 2026**, advancing controllable generation across camera, temporal, spatial, and action dynamics.

Selected Publications

Yu Yuan, Jianhao Yuan, Xijun Wang, Daiqing Li, Liu He, Lu Ling, Stanley H. Chan.
OptiWorld: Optimal Control for Video World Generation under Physical Constraints
2026 [Under Review]

Yu Yuan, Tharindu Wickremasinghe, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan.
SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation
CVPR 2026

Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, Bole Ma, Stanley H. Chan.
NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics.
ICLR 2026

Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, Stanley H. Chan.
Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis.
CVPR 2025 [Highlight & Demo]

Yu Yuan, Yiheng Chi, Xingguang Zhang, Stanley H. Chan.
iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures.
ICIP 2025

Xingguang Zhang, Nicholas Chimitt, Xijun Wang, **Yu Yuan**, Stanley H. Chan.
Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation.
CVPR 2025 [Highlight]

Lanqing Guo*, Xijun Wang*, Minchul Kim*, **Yu Yuan**, Wes Robbins, Xingguang Zhang, Stanley H. Chan, Zhangyang Wang, Xiaoming Liu. ***Beyond Clean Pixels: Interdisciplinary Lessons from Turbulent Long-Range Face Recognition.***
2025 [Under Review]

Joonyeoup Kim, **Yu Yuan**, Xingguang Zhang, Xijun Wang, Stanley H. Chan.
Astrophotography Turbulence Mitigation via Generative Models.
ICIP 2025

Lindong Wang, Hongya Tuo, **Yu Yuan**, Henry Leung, Zhongliang Jing.
RCMixer: Radar-Camera Fusion Based on Vision Transformer for Robust Object Detection.
JVCI 2024

Yu Yuan, Jiaqi Wu, Lindong Wang, Zhongliang Jing, Henry Leung, Shuyuan Zhu, Han Pan.
Learning to Kindle the Starlight.
arXiv 2022

Yu Yuan, Jiaqi Wu, Zhongliang Jing, Henry Leung, Han Pan.
Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention.
arXiv 2022

Patent

Trajectory resolving and registering method based on autonomous underwater robot inertial navigation and ultra-short baseline positioning sensor. CN 114993313 A.

Awards & Competitions

Ivy League Scholarship for Exceptional Students (Innovation)	Top 1%
Outstanding Graduate of Shanghai Jiao Tong University	Top 10%
Shanghai International Creators Competition: Special Competition of Drones	First Prize
DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car	National Second Place

Other Activities

Teaching Assistant, Automatic Control Theory, Shanghai Jiao Tong University

Workshop Organizer, WACV GAIP 2026

Reviewer, CVPR, ICCV, ICLR, NeurIPS, WACV, CVPRW, ICIP, ICASSP

IEEE Graduate Student Member

Skills

Languages: English (Professional Working Proficiency), Chinese (Native)

Programming: Python, PyTorch, HTML, JavaScript, Matlab, C/C++ (Basic)

Technologies: Diffusers, Transformers, Post-training, 3DGS, CUDA, OpenCV, NumPy, Git, LaTeX, Linux, Slurm, Markdown, Gradio, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere

Others: Photography, Fishing, Drones, Rowing